

Self-Organizing Federated Learning in Collective Adaptive Systems

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Abstract—Collective Adaptive Systems (CAS) require learning mechanisms that are decentralized, privacy-preserving, and robust to heterogeneity, mobility, and failures. This research investigates how self-organizing learning methods can support Federated Learning (FL) in such settings, enabling collaboration structures to emerge and adapt from local device interactions rather than being fixed a priori. The proposed approaches exploit spatial proximity and model similarity to organize devices into adaptive learning regions under non-IID data. Experimental results show improved accuracy, resilience, and efficiency over standard FL baselines, while supporting mobility and resource constraints.

Index Terms—Collective Adaptive Systems, Federated Learning, Self-Organization, Non-IID Data.

I. MOTIVATION AND CHALLENGES

CAS are large-scale ensembles of situated computational entities that interact locally and adapt to changing environmental and operational conditions [1]. Their distinctive feature is the emergence of system-level behaviour from the coordination of many components, which together provide capabilities that would not be available to isolated elements [2]. This makes CAS a natural abstraction for several contemporary computing scenarios, including smart cities, the Internet of Things, pervasive sensing infrastructures, and fleets of autonomous robots, where many heterogeneous devices cooperate under changing environmental and operational conditions [1], [3].

In these settings, intelligence is increasingly expected to emerge from the collective rather than from a single central component. Devices continuously sense local phenomena, exchange information with nearby peers, and adapt their behaviour according to partial and situated knowledge. Learning is therefore a key enabler for CAS: it allows the system to exploit distributed experience, improve its behaviour over time, and support cooperative decision-making in environments that are too dynamic, large, or uncertain to be fully modelled a priori. However, bringing learning capabilities into CAS requires moving beyond conventional centralized machine learning pipelines, which assume stable infrastructures, globally available data, and centralized control.

Learning in CAS must instead cope with a combination of intertwined challenges: (i) decentralized control; (ii) heterogeneous and spatially situated data; (iii) dynamic participation and mobility; and (iv) privacy constraints. These aspects are

not independent. Rather, they jointly shape the way knowledge can be produced, shared, and exploited by the collective.

Decentralization is a first fundamental issue. CAS are typically composed of devices that interact locally and opportunistically, without a reliable global view of the system. As a consequence, learning cannot rely on a fixed coordinator or on the assumption that all relevant information can be collected in one place. Each device only observes a partial view of the environment and of the collective state, while global behaviour must emerge from local exchanges. This raises the problem of how to structure collaboration among devices, how to aggregate knowledge through peer-to-peer interactions, and how to maintain robustness when communication is intermittent, asymmetric, or affected by failures.

At the same time, data in CAS are inherently heterogeneous. Each device observes the environment from a specific position, with its own sensing capabilities, operating conditions, and interaction history. This naturally leads to non-independent and identically distributed (non-IID) data. Spatial locality further reinforces this phenomenon: nearby devices tend to experience similar conditions, whereas distant devices may be exposed to substantially different phenomena [4]. Therefore, heterogeneity should not only be treated as a limitation to be mitigated, but also as a source of contextual information. Learning mechanisms for CAS should be able to exploit relations such as proximity, neighbourhood structure, and environmental similarity to decide which devices should collaborate and how their knowledge should be combined [5].

These difficulties are amplified by the dynamic nature of CAS. Devices may join or leave the system, communication links may appear and disappear, and mobile entities may change their position with respect to the phenomena being observed. Such changes affect both the topology of interactions and the data distributions underlying the learning process. Hence, learning in CAS cannot be seen as a one-shot training phase followed by deployment; it must be continuously adaptive. The system should be able to reorganize collaborations over time, recover from transient perturbations, and remain effective under churn, mobility, and evolving environmental conditions.

Finally, learning in CAS must account for privacy. Since devices often collect data about people, places, and physical processes, transferring raw observations to a central location

may be undesirable or even infeasible.

II. CONTRIBUTION AND OBJECTIVE

This research investigates how learning can be engineered in CAS by combining decentralized and clustered FL [6], [7] with self-organizing coordination mechanisms. Its first contribution is the characterization of the CAS properties that make conventional machine learning (ML) approaches difficult to apply, including decentralized control, spatially heterogeneous data, mobility, dynamic participation, and privacy constraints. Within this space, FL is identified as a promising starting point, since it enables devices to train collaboratively without centralising raw data.

A core objective of the research [5], [8] is to relate decentralized FL to Aggregate Computing (AC) [9], a coordination paradigm for resilient and self-* systems. AC has increasingly been combined with ML techniques due to its ability to express adaptive collective behaviours under failures, mobility, and changing network conditions [10]–[12]. In this perspective, AC supports the specification of FL processes at the collective level, allowing the federation structure to emerge from local device interactions rather than being fixed a priori, for instance through leader election and adaptive neighbourhood formation [13].

The research also contributes tools and experimental artefacts. ProFed [14] is introduced as a benchmark for non-IID FL scenarios where devices observe spatially correlated data, while Phydels [15] provides a Python-hosted AC language designed to interoperate with mainstream ML libraries such as PyTorch, overcoming the practical limitations of existing AC implementations based on non-AI-oriented ecosystems [16], [17].

Finally, the research analyses how device mobility affects learning when data distributions evolve over time, and explores Continual Learning techniques to improve robustness under such changes [18]. It further investigates lightweight collaboration strategies based on Random Network Distillation (RND) [19] as a novelty signal for clustering devices that should cooperate within the federation, mitigating accuracy degradation caused by non-IID data [20]. In parallel, neural network sparsification is integrated to support learning on resource-constrained devices, studying the trade-off between model compression and predictive performance [21].

III. METHODOLOGY AND RESULTS

The research follows an experimental and constructive methodology, combining the design of self-organizing learning algorithms with simulation-based evaluation. The proposed approaches model CAS as spatially distributed networks of devices executing local computation rounds, where learning and coordination are both driven by neighbourhood interactions. AC abstractions, such as leader election, information collection, and information spreading, are used to express decentralized FL processes as collective behaviours, enabling devices to form adaptive learning regions without relying on a fixed central server.

The resulting methods have been evaluated on proximity-based non-IID scenarios generated through ProFed, using standard vision datasets such as MNIST, FashionMNIST, EMNIST, CIFAR-100, and UTKFace. Results show that field-based and proximity-aware federations can match centralized FL in IID settings, while improving accuracy over FedAvg, FedProx, Scaffold, and clustered baselines under spatially non-IID data. Experiments also indicate that self-organizing architectures improve robustness to aggregator failures and support stable federation formation under mobility. Finally, further studies show that lightweight model-similarity signals and neural network sparsification can reduce communication and computational costs while preserving competitive accuracy, making the approach suitable for resource-constrained CAS.

IV. FUTURE WORK AND RESEARCH PLAN

Future work will focus on strengthening the evaluation and adaptability of self-organizing FL for realistic CAS scenarios. A first direction is the development of benchmarks based on real-world spatial data, such as traffic, mobility, or environmental traces. Current benchmarks rarely capture at the same time spatial data heterogeneity, dynamic participation, mobility, and resource constraints, making it difficult to compare algorithms under realistic conditions.

A second direction is to study FL more explicitly as a self-adaptive system. Existing work in this area [22], [23] mostly focuses on adaptations such as client selection, while architectural adaptations, e.g., switching between centralized, hierarchical, and decentralized aggregation, are often considered too disruptive to be dynamically activated or deactivated. Field-based coordination offers a promising way to revisit this limitation, since it can express architecture reconfiguration as an emergent collective behaviour driven by local interactions [10].

Finally, the research will investigate cheaper proxy metrics for estimating neighbour similarity and deciding which devices should collaborate, reducing the cost of current loss- or model-based comparisons. This will be complemented by experiments on deployment-oriented constraints, including communication frequency, energy consumption, partial participation, and unreliable connectivity, to better assess the feasibility of self-organizing federations in real CAS.

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